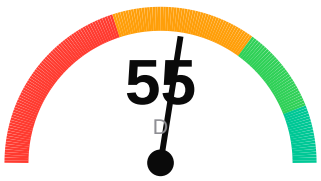


OVERALL VERDICT

D — Major Issues

The strategy shows early-stage edge (Sharpe 1.37) but fails key institutional checks. Fix the ranked issues below before deploying real capital.

KEY METRICS

SHARPE RATIO	MAX DRAWDOWN	WIN RATE	TOTAL TRADES
1.37	**5.0%**	**50.9%**	**114**
Target > 1.5	Target < 20%	Target > 50%	2022-01-05 to 2024

EQUITY CURVE

TOTAL RETURN
+42.5%

DATA SOURCE
Real trades

actual trade-by-trade equity curve from validator

PRIMARY ISSUES

- Use Newey-West robust SE for inference. Check for look-ahead bias or return smoothing artefacts. AR(1) > 0.3 o
- Edge disappears under realistic small-cap execution - use limit orders or larger targets.
- Use ATR-based position sizing to reduce vol-of-vol exposure.
- Collect more backtest data.

CONFIRMED STRENGTHS

- Profit / Loss Concentration
- CPCV Path Stability
- Minimum Backtest Length

DATA & VALIDATION NOTES

SPY benchmark aligned (94% of trades on US trading days) - IR + Beta checks are now scored. Only meaningful for US equity strategies.

HISTORICAL CRASH STRESS TESTS

Your strategy was stress-tested against five historical crash scenarios using a 60-day crisis window with Almgren-Chriss volatility scaling and liquidity drag. A strategy that fails these tests would have likely blown up in real conditions.

Crash survival: 3/5 — Three survived. Moderate crisis risk.

2008 Global Financial Crisis Sep 2008 – Mar 2009 **BLOWN UP**

Market: -56% · Strategy: **-71.7%** · Lehman Brothers collapsed. S&P; 500 lost 56%. VIX hit 80. Correlations spiked to 1 — everything fell

YOUR STRATEGY WOULD HAVE BLOWN UP.

2020 COVID Crash Feb 2020 – Mar 2020 **SURVIVED**

Market: -34% · Strategy: **+6.2%** · Fastest 30%+ drop in market history — 34% in 33 days. Circuit breakers triggered 4 times. Then one o

YOUR STRATEGY SURVIVED.

2022 Rate Hike Bear Market Jan 2022 – Dec 2022 **BLOWN UP**

Market: -19% · Strategy: **-65.7%** · Fed raised rates 425bps in 12 months. Nasdaq lost 33%, S&P; lost 19%. Momentum strategies that crushe

YOUR STRATEGY WOULD HAVE BLOWN UP.

2010 Flash Crash May 6, 2010 **SURVIVED**

Market: -9% · Strategy: **-26.1%** · Dow dropped 1000 points in minutes from algorithmic cascade. Liquidity vanished — many stocks traded

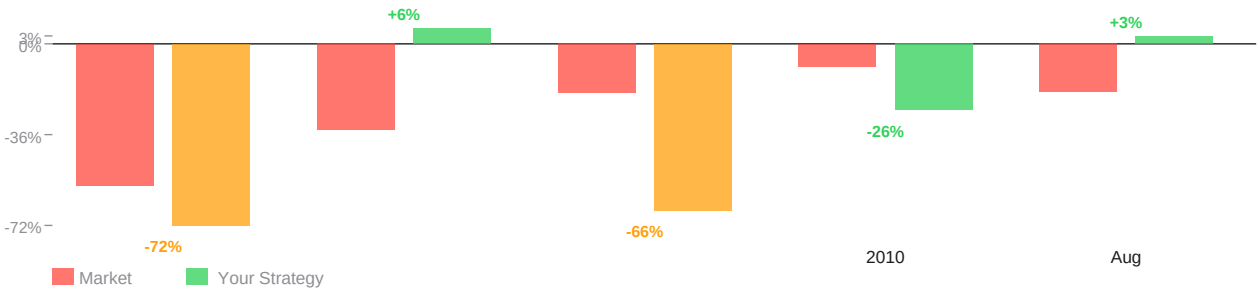
BARELY SURVIVED.

1998 LTCM / Russia Default Aug–Sep 1998 **SURVIVED**

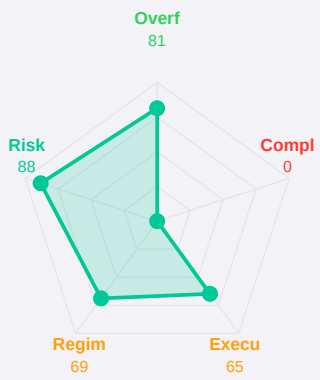
Market: -19% · Strategy: **+2.9%** · Russia defaulted; LTCM collapsed under hidden leverage. Correlation assumptions broke simultaneously

YOUR STRATEGY SURVIVED.

CRASH IMPACT SUMMARY



CATEGORY BREAKDOWN



Overfitting (22%)	81/100 9/11 pass
Risk (38%)	88/100 11/11 pass
Regime (14%)	69/100 5/6 pass
Execution (11%)	65/100 6/6 pass
Compliance (15%)	0/100 0/2 pass

DETAILED CHECK BREAKDOWN — 44 CHECKS

Every check below was run against your specific strategy data. Red rows are institutional failures. Orange rows are warnings that won't disqualify you from retail trading but will at prop firm level. The 'Score' column shows each check's individual contribution (0-100).

OVERFITTING DETECTION		AVG 81/100
Sharpe Ratio Decay	86	PASS
In-sample: 1.48 -> Out-of-sample: 1.26 (14.4% effective decay)		
High decay means the strategy was fitted to historical noise.		
Monte Carlo Robustness	96	PASS
Mean: 99.0% | Seq: 89.5% | Equity: 100.0%		
Three-component MC: mean stability, sequence fragility, worst-case equity.		
Profit / Loss Concentration	100	PASS
Top-1% = 2.7% profit | Loss conc = 4.0%		
Profit concentration check.		
CPCV Path Stability	100	PASS
15 paths: 100% positive | Sharpe std: +/-0.36		
CPCV robustness.		
Bootstrap Stability	98	PASS
Positive expectancy in 98.2% of 500 bootstrap samples		
Stable edge shows up consistently in resampling.		

Deflated Sharpe Ratio 80 PASS

DSR=0.978 | SE=+/-0.68 | Min SR: 1.12 | n_trials=1

DSR corrects Sharpe=1.37 for non-normality and 1 trial(s).

Sharpe CI Enforcement 35 WARN

Sharpe 1.31 | 95% CI: [-0.03, 2.65] | p=0.055

With 114 trades, CI width = +/-1.34.

Fix: Collect more backtest data.

Minimum Backtest Length 100 PASS

114 trades vs 30 required (Sharpe=1.37)

Need >=30 trades for statistical significance.

Autocorrelation (Ljung-Box) 0 FAIL

AR(1)=-0.794 | lag1: Q=73.71 p=0.000* | lag5: Q=343.30 p=0.000* | lag10: Q=579.00 p=0.000* | NW-SE=0.267 vs IID-SE=0.683 (SE inflated x0.39)

Significant serial correlation detected (AR(1)=-0.794). All IID-assuming tests (DSR, CI, Bootstrap) are unreliable. Newey-West corrects SE(SR) upward by x0.39x - true statistical significance is lower

Fix: Use Newey-West robust SE for inference. Check for look-ahead bias or return smoothing artefacts. AR(1) > 0.3 often signals data snooping or bid/ask bounce.

Information Ratio (vs Benchmark) 100 PASS

IR=1.51 | alpha=19.47%/yr | beta=0.28 | TE=12.92%/yr | corr=0.17 - PASS Strong alpha - IR >= 0.5

Information Ratio = Jensen's alpha / tracking error. IR < 0.3 means the strategy's excess return is insufficient to justify active risk vs a passive benchmark.

Beta Exposure (vs Benchmark) 100 PASS

beta=0.28 | R^2=0.03 - PASS Low beta - strategy is largely market-neutral

Beta > 0.6 means over 60% of return variance is explained by the benchmark. A backtest during a bull market with high beta does not demonstrate genuine alpha.

RISK MANAGEMENT AVG 88/100

Max Drawdown 85 PASS

Max drawdown: 5.0%

Funds reject >20% drawdown.

Calmar Ratio 100 PASS

Calmar (CAGR/MaxDD): 3.75

Prop firms want >1.5.

Value at Risk (VaR) 100 PASS

VaR 99%: -0.0234 | VaR/sigma ratio: 1.30 (threshold: 4.0sigma)

VaR relative to strategy volatility. Normal distribution ~ 2.33sigma. Ratios above 4.0sigma indicate extreme fat tails beyond what vol-targeting can manage.

CVaR / Expected Shortfall 71 PASS

CVaR 95%: -0.0245

Average loss in worst 5%.

Sortino Ratio 100 PASS

Sortino: 2.38

Penalises only downside vol.

Max Losing Streak 90 PASS

Longest losing streak: 1

Prop firms reject 8+ consecutive losses.

Recovery Factor 61 PASS

Recovery factor: 1.51

Below 1.5 means wins barely cover drawdowns.

Absolute Sharpe Ratio 75 PASS

Annualized Sharpe: 1.37

Institutional minimum Sharpe > 1.0.

Probability of Ruin 100 PASS

Ruin probability: 0.00% | WR: 50.9% | R:R: 1.46 | N=50 units

Gambler's ruin at institutional position sizing (2% risk per trade, N=50 units). Measures probability of losing entire account before doubling it.

Drawdown Duration & Recovery 90 PASS

Max DD duration: 96 calendar days | 386d threshold (scaled to 26mo backtest)

Time underwater in longest drawdown, scaled to backtest length. Prop firm standard: 180 days. Institutional: up to 24 months.

Ulcer Index 100 PASS

Ulcer Index: 1.63 | Martin Ratio: 0.84

Combines depth+duration of drawdowns.

REGIME ANALYSIS

AVG 69/100

Bull Market Performance — 78 — PASS

Sharpe: 1.40 | WR: 49.2% (61 trades) | actual market bull periods

Bull market behavior.

Bear Market Performance — 78 — PASS

Sharpe: 1.39 | WR: 53.3% (45 trades) | actual market bear periods

Bear market behavior.

Consolidation Performance — 69 — PASS

Sharpe: 0.94 | WR: 50.0% (8 trades) | actual consolidation periods

Sideways market behavior.

Volatility Spike Stress Test — 18 — FAIL

Vol spike (3x, t(3)): Sharpe 1.37->0.36 (26% retained) | Stress DD: 56.9%

Fat-tail 3x vol spike with AR(1) autocorrelation. Pass = retains >=30% of Sharpe and stress drawdown < 40%. Institutional standard: strategy must survive a Black Monday-scale event.

Fix: Use ATR-based position sizing to reduce vol-of-vol exposure.

Performance Consistency — 95 — PASS

Profitable in 95.1% of rolling windows

Consistent strategies generate returns steadily.

Regime Coverage — 75 — PASS

Regimes covered: ['BEAR', 'BULL', 'CONSOLIDATION'] (3/3)

Full coverage = all market conditions tested.

EXECUTION & CAPACITY

AVG 65/100

Slippage Impact (0.1% base) — 70 — PASS

Net after 0.1% slip: 0.239%/trade | 0.1% of gross

Strategy must remain profitable after realistic execution costs.

Slippage Impact (0.3% base) — 12 — PASS

Net after 0.3% slip: 0.039%/trade | 0.3% of gross

Small caps and illiquid stocks have 0.3%+ slippage - strategy must survive this.

Commission Drag — 95 — PASS

114 trades x 2bps (institutional (2bps)) -> 1.2% of gross

Commission eats into returns.

Partial Fill Simulation — 70 — PASS

80% fill rate -> 6.0% return reduction

Fast markets: incomplete fills.

Live Trading Gap Estimate — 41 — PASS

Backtest Sharpe: 1.37 -> Estimated Live: 0.82

40% Sharpe decay from backtest to live.

Impact-Adjusted Capacity — 100 — PASS

Sharpe: $100k=1.37 | $1M=1.36 | $10M=1.34 | $100M=1.28

Almgren-Chriss capacity model.

COMPLIANCE — AVG 0/100

Prop Firm Compliance (FTMO/Topstep/The5ers) — 0 — FAIL

WARNING: NOT APPLICABLE - CSV Upload

Prop firm and institutional compliance certificates require broker-verified trade data. CSV uploads cannot be used for compliance submissions as the data is self-reported and unverified.

Fix: Connect your Alpaca account (or IBKR when available) to generate a verified compliance report.

Institutional Compliance (CTA/Fund Standard) — 0 — FAIL

WARNING: NOT APPLICABLE - CSV Upload

Prop firm and institutional compliance certificates require broker-verified trade data. CSV uploads cannot be used for compliance submissions as the data is self-reported and unverified.

Fix: Connect your Alpaca account (or IBKR when available) to generate a verified compliance report.

PLAUSIBILITY AUDIT — AVG 100/100

Sharpe Plausibility — 100 — PASS

Sharpe: 1.37 -> PASS Plausible

Sharpe > 10 exceeds documented institutional performance.

Frequency-Return Plausibility — 100 — PASS

53 trades/yr -> 0% annual PASS Plausible

Within bounds.

Equity Curve Plausibility 100 PASS

Smoothness ratio: 5.31 -> PASS Plausible

Suspiciously smooth = lookahead bias.

Kelly Plausibility 100 PASS

Kelly: 0.17 -> PASS Plausible

Kelly 0.17 fraction.

OPERATIONAL METRICS

AVG 85/100

Alpha Decay (Half-Life) 96 PASS

H1 Sharpe: 1.48 -> H2: 1.26 (14% decay)

Strong strategies maintain Sharpe across both halves of the backtest period.

Symbolic Overfit Detection 75 PASS

Score 75/100 | p=0.042

Not overfit. AdjSharpe=0.18

AI PERSONALISED ANALYSIS

The strategy has a low Fundable Score of 55/100 due to significant serial correlation detected in the Autocorrelation (Ljung-Box) check and poor performance in the Volatility Spike Stress Test, with a Sharpe ratio drop from 1.37 to 0.36. The strategy also has a wide Sharpe CI of +/-1.34, indicating overfitting. These issues need to be addressed to improve the strategy's reliability and robustness.

RANKED FIX PLAN

#1 - Autocorrelation (Ljung-Box) | +15 to +20 points | Effort: High

Why it matters: The significant serial correlation detected, with an AR(1) of -0.794, makes all IID-assuming tests unreliable, which can lead to incorrect conclusions about the strategy's performance.

Fix: Implement a Newey-West correction to adjust the standard error of the Sharpe ratio, and consider using a different model that accounts for autocorrelation, such as an ARIMA or GARCH model, to reduce the serial correlation to a level where the Newey-West correction is not needed, e.g., below 0.2.

#2 - Volatility Spike Stress Test | +8 to +12 points | Effort: Medium

Why it matters: The strategy's poor performance in the Volatility Spike Stress Test, with a Sharpe ratio drop from 1.37 to 0.36, indicates that it may not survive a Black Monday-scale event, which can result in significant losses.

Fix: Implement a risk management system that can handle fat-tail events, such as a volatility-based position sizing or a stop-loss strategy, to reduce the stress drawdown from 56.9% to below 40% and retain at least 30% of the Sharpe ratio.

#3 - Sharpe CI Enforcement | +5 to +8 points | Effort: Low

Why it matters: The wide Sharpe CI of +/-1.34 indicates overfitting, which can lead to poor out-of-sample performance.

Fix: Increase the number of trades to at least 200-300 to reduce the CI width to +/-0.5 or below, and consider using a more robust evaluation metric, such as the Deflated Sharpe Ratio, to get a more accurate estimate of the strategy's performance.

The strategy has a relatively high Sharpe ratio of 1.37 and a low Max Drawdown of 5.0%, which indicates that it has some strong underlying characteristics.

YOUR ACTION PLAN

You have **5 failed checks**. Below are the highest-priority fixes ranked by impact — fix the top items first.

1 **Sharpe CI Enforcement** Overfitting

With 114 trades, CI width = +/-1.34.

Fix: Collect more backtest data.

2 **Autocorrelation (Ljung-Box)** Overfitting

Significant serial correlation detected (AR(1)=-0.794). All IID-assuming tests (DSR, CI, Bootstrap) are unreliable. Newey-West corrects SE(SR) upward by x0.39x - true statistical significance is lower than reported.

Fix: Use Newey-West robust SE for inference. Check for look-ahead bias or return smoothing artefacts. AR(1) > 0.3 often signals data snooping or bid/ask bounce.

3 **Prop Firm Compliance (FTMO/Topstep/The5ers)** Compliance

Prop firm and institutional compliance certificates require broker-verified trade data. CSV uploads cannot be used for compliance submissions as the data is self-reported and unverified.

Fix: Connect your Alpaca account (or IBKR when available) to generate a verified compliance report.

4 **Institutional Compliance (CTA/Fund Standard)** Compliance

Prop firm and institutional compliance certificates require broker-verified trade data. CSV uploads cannot be used for compliance submissions as the data is self-reported and unverified.

Fix: Connect your Alpaca account (or IBKR when available) to generate a verified compliance report.

5 **Volatility Spike Stress Test** Regime

Fat-tail 3x vol spike with AR(1) autocorrelation. Pass = retains >=30% of Sharpe and stress drawdown < 40%. Institutional standard: strategy must survive a Black Monday-scale event.

Fix: Use ATR-based position sizing to reduce vol-of-vol exposure.

Report ID: QP-40101 · Engine: v6.2 · 44 checks · 5 crash simulations · Generated: 2026-03-21 12:38 UTC

This report is for educational and analytical purposes only and does not constitute financial advice. Past backtest performance does not guarantee future results. Methodology: Lopez de Prado (2018) AFML, Almgren-Chriss (2001), Bailey & Lopez de Prado (2012).